

CROI: Final Design Report | Pulsometer Technology

Introduction & Objectives

A pulseometer is a wearable device to measure a user's real-time heart rate. Pulseometers are commonplace for use in both medical settings and at-home patient care. Commercial pulseometers measure heart rate (bpm) and peripheral blood oxygen saturation (SpO2). These measurements are useful for everyday health monitoring, but are particularly important for patients with conditions affecting the lungs or heart - like COPD, COVID-19, or heart disease.

This Final Report gives the underlying theory for this device and outlines our final design, with the goal that it could be marketed as a commercially viable finger clip pulseometer. It incorporates real-time heart rate monitoring with a user-friendly design and visual display. Using the principles of photoplethysmography, we are able to detect the subtle flushing that occurs when blood flows through capillary arteries near the surface of the skin. Then, using various signal processing methods, we turn this information into a heart rate. This device does not measure oxygen saturation as that was outside the scope of this course, but could be considered in future iterations of this project.

Economic & Market Analysis

The global market for wearable healthcare devices is projected to increase by \$75.98 billion by 2030, representing a 67.76% increase over its 2025 value. Over this forecast period, the market is expected to grow at a compound annual growth rate (CAGR) of 10.9%. This shows the demand for wearable healthcare devices will continue to grow. This market includes devices that measure pulse, blood-oxygen levels, glucose, insulin, and more with the purpose of focusing on this market because there isn't an explicit pulsometer market, only a pulse oximeter market. For the SWOT analysis, we have identified the following:

- **Strength:** Our product only has one feature and so our focus is to have the pulse readings more accurate than other products.
- **Weakness:** Since we are using an arduino, the product will be more clunky to wear.
- **Opportunity:** general health awareness amongst people is rising, making it more likely our product will sell.
- **Threat:** Most pulsometers on the market also function as an oximeter, making our product have less features than competitors.

To better understand where our product would sit in the market, we looked at competitors that offer heart rate monitoring as part of their product. The smartwatch category (Apple, Samsung, Fitbit) typically offers heart rate, blood oxygen, sleep tracking, and other fitness features at a price range of roughly \$65 to \$799. The dedicated pulse oximeter category (Masimo, Medtronic, Wellue) offers heart rate and blood oxygen at a price range of roughly \$20 to \$700. Across all six competitors we looked at, the average price range is approximately \$148 to \$522. In our presentation, we simply listed our competitors as \$50 since this was what we found in our local CVS.

Looking at the bill of materials for our own product, the component cost is significantly lower than what is on the market. The IR LED is roughly \$0.90, the photodiode is \$1.40, the resistors total \$0.25, the capacitors total \$0.21, the op-amps total \$0.64, the microcontroller is between \$5 and \$15 depending on whether we use the Arduino Uno or a smaller microcontroller, and the mini LCD is roughly \$5. The total component cost is between \$13.40 and \$23.40. Even after accounting for the 3D printed housing, assembly, and a reasonable markup, our product could be priced well below the competitor average while still maintaining margin. The tradeoff is that our product currently only measures heart rate, not blood oxygen, and is bulkier than a smartwatch.

Methodology & Theory

Our pulseometer, like commercial products, is based on photoplethysmography (PPG). PPG is an optical technique based on the Beer-Lambert law, and it can be used to measure physiologic information about a patient. When a patient's skin is illuminated with specific wavelengths of light, the absorbance fluctuates in accordance with the flow of blood through that area of tissue. If one can measure this real-time absorbance, it is possible to extract information about not only heart rate, but blood oxygen saturation, respiratory rate, and even sleep quality.

Our project will use transmittance PPG, where the light source is positioned opposite to the light collecting device, so the signal must travel through the appendage (fingertip) as demonstrated in the figure. In order to pick up small absorbance changes in pulsatile artery blood, we need to understand the optical properties of tissue and blood. Considering we are using transmittance PPG, light must be able to travel through the fingertip without getting completely absorbed or scattered out. Light scattering in tissue can be modeled with a power law relationship:

$$\mu_s' \propto \lambda^{-b}$$

Where:

μ_s' = scattering coefficient [1/length]

λ = wavelength of light

b = scattering "strength" [dimensionless]

Depending on the tissue, b could be anywhere between 0.7-1.5. It is clear that as wavelength decreases, scattering increases, and we see a similar trend for absorption. Absorption follows the Beer-Lambert law:

$$A = \epsilon Lc$$

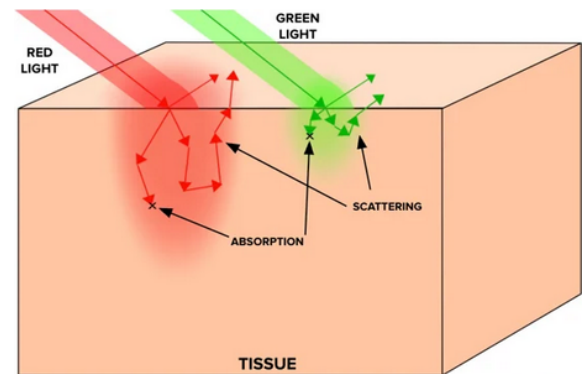
Which can be re-expressed as:

$$\frac{I}{I_0} = 10^{-A} = 10^{-\epsilon Lc} = e^{-\mu_a A}$$

Where:

A = Absorbance [dimensionless]

ϵ = molar extinction coefficient [1 / concentration*length]



L = path length

c = concentration

I_0 = Incident intensity [power/area]

I = transmitted intensity [power/area]

μ_a = absorption coefficient [1/length]

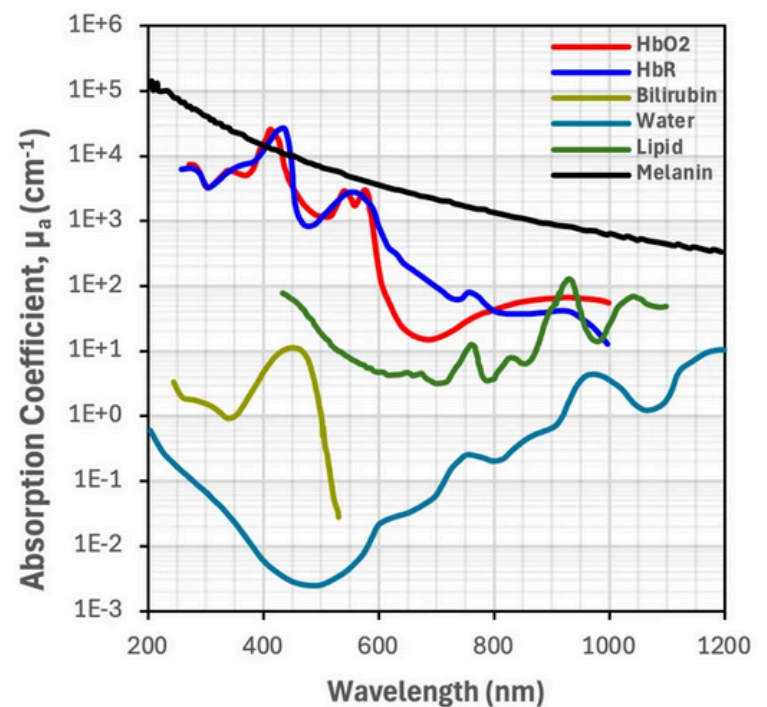
Re-arranging the expressions above we see that $\mu_a \propto c\epsilon$. However, in tissue samples this becomes slightly more complicated, as the Beer-Lambert law assumes the presence of only one absorber. In skin and tissue there are many absorbers, so we can express this with a sum:

$$\mu_a \propto \sum_i c_i \epsilon_i$$

Each of these absorbers has slightly different molar extinction coefficients (melanin and other chromophores in the skin exhibit strong light absorption in the 200-500nm range, whereas water and lipids become more significant past 800nm). Absorbance is not as simple as scattering, but in general molar extinction coefficients for relevant organic molecules in the skin become stronger at lower wavelengths. Therefore if the goal is for light to travel all the way through tissue, it is more helpful to use a higher wavelength of light. It must still be a wavelength that can pick up on vascular activity; that is, it must be absorbed or scattered just enough so that we can detect a change when blood flow is occurring.

Deoxygenated and oxygenated hemoglobin absorb some wavelengths light equally (red-NIR), but some wavelengths differently, and this contrast allows us to collect

absorbance data. Infrared light (850-1000nm) is more absorbed by oxygenated hemoglobin, while deoxygenated hemoglobin has stronger absorption of red light (650–750nm). In commercial devices, the difference between these absorbance ranges allows one to measure blood oxygen saturation. However, for the purpose of this device we will therefore be using infrared light, as it will have enough penetration depth into the tissue to be detected with transmittance PPG while also being sufficiently scattered / absorbed by oxygenated blood for us to detect it.

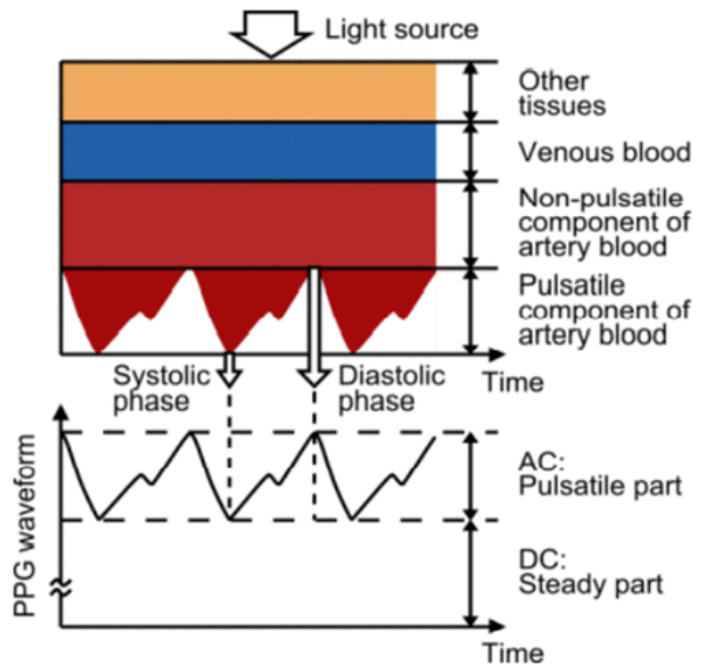


Engineering Design Challenges

Challenges in producing an effective & accurate pulsometer relate to understanding the interaction of light with human skin tissue, small signals with a large amount of background noise, user interaction with the equipment.

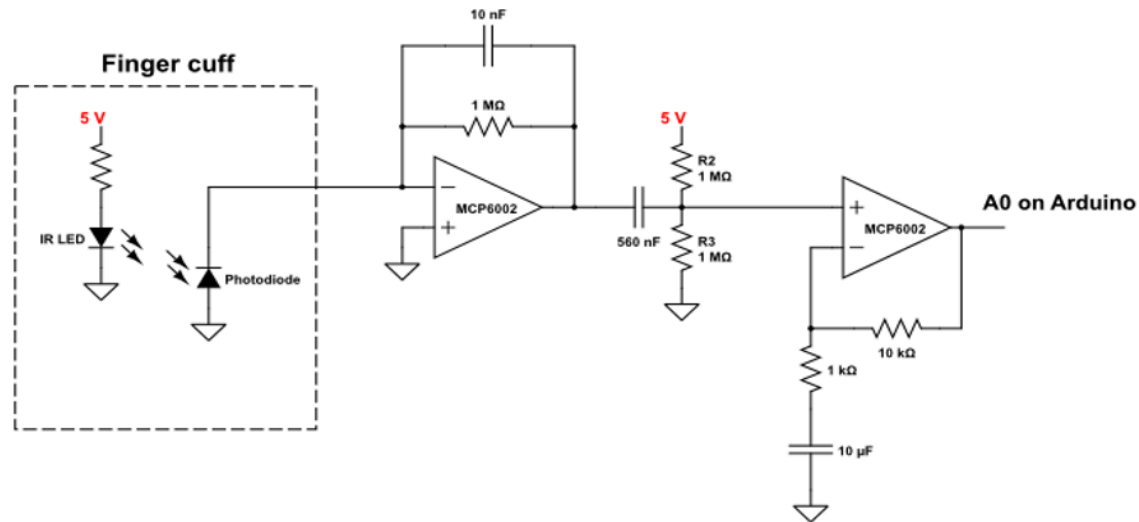
The focus point for our device is the finger, yet even the composition of the finger is very complex and tissue properties vary widely from person to person. In our device, light must travel through every skin layer (epidermis, dermis, and hypodermis), then through muscle, fascia, nerve tissue, and other connective tissue. Our device should be able to function regardless of skin color (as melanin content can change refractive behavior of light), fat content, or muscle content. While the patient is using the device, it should still be able to function under differing conditions of fingertip perspiration, slight finger movement, or changing external lighting conditions. We later see that this design challenge: user movement and finger positioning, is the greatest challenge. We plan to account for these differences by using wavelengths of light that work regardless of skin absorbers (like melanin) and target the absorption spectra of oxygenated blood. We try to account for friendly user design by using a finger clip that will block out external light sources and will grip tight enough to not be affected by user movement, but not so tight that it cuts off blood flow.

A challenge we faced in CBE 3300A when creating a rudimentary pulsometer is making a small signal amplified and removing background signal and noise. The flushing of the skin that we are trying to detect is very minor, and creating a device to measure these slight changes could be likened to finding a needle in a haystack. There are many sources of background signal and noise for PPG, but absorption from other tissues, background light, and user movement were the largest. However, these factors generally do not change for an individual user while they are using the device, so it tends to result in a baseline DC signal that we have to filter out. What is important is making sure the signal is able to stand out after we remove everything else. We plan to deal with this by implementing new circuit design and signal processing techniques.



Circuit Design & Signal Processing

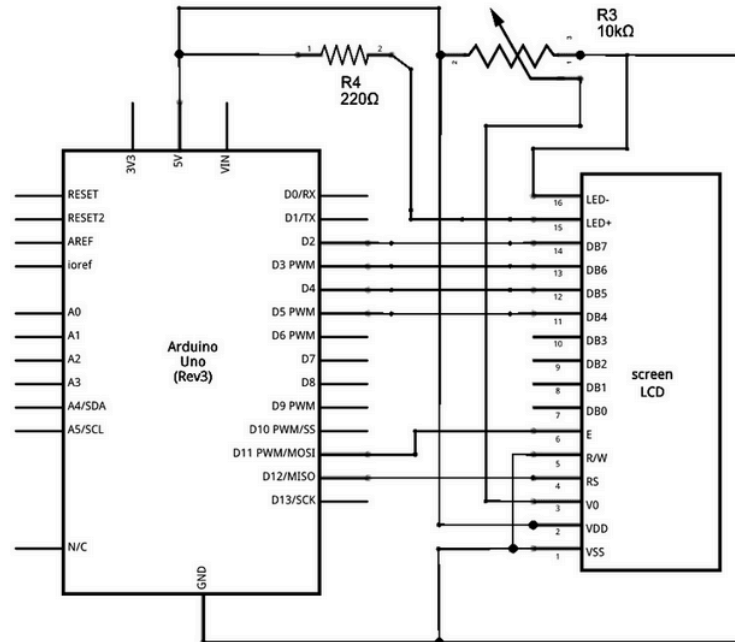
Most current circuit design shown below:



This circuit is made up of four components. Going left to right, the first component is the finger cuff, this is where the patient interacts with the device. The finger cuff will be in the shape of a finger clip, with the IR LED and the photodiode positioned on opposite sides of the finger. The photodiode converts light into electrical current with a high sensitivity and linearity, which is helpful for the small signals we are working with. The second component is the transimpedance amplifier (TIA). The current from the photodiode flows to the negative terminal of the op-amp and through the gain-setting resistor. This creates a voltage at the output of the op-amp that is proportional to the input current - the capacitor helps reduce noise. Next is a simple highpass filter; we are removing the DC baseline signal that comes from the natural absorbance and scattering from the tissue. The last component is a non-inverting gain op-amp. This amplifies the signal by a factor of $R2/R1$ (the ratio of the resistor in the feedback loop to the resistor connected to ground). The capacitor in this stage blocks the DC signal that we added in the previous stage, amplifying only the AC signal that we care about.

The goal of this whole process is to isolate the signal we care about (absorbance from oxygenated hemoglobin) and amplify it while removing background signal / noise. In this circuit there is already signal processing with multiple built-in filters, but there is more signal processing that has to happen in arduino. In order to give a heart rate in bpm, it is necessary to real-time peak detection on the signal. Explained further below.

We also added an LCD screen to read the BPM from the pulseometer and portray a heart or skull depending on the pulseometer reading. The screen circuitry is shown below.



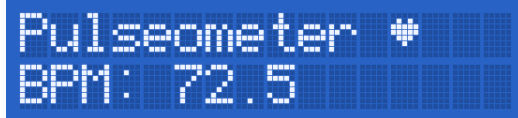
Product's Code

First we have the window, which is a rolling buffer that stores the last 4 seconds of sensor readings (80 samples) so the program always has a fresh chunk of recent data to analyze.

Then we have the peak detection. Every 2 seconds the program scans through all the samples in the window and checks each one against three rules. The first rule is that the sample has to be higher than the samples directly before and after it, meaning it sits at a local high point. The second rule is that the peak has to be at least 10 samples (half a second) away from the previous accepted peak, which keeps a single heartbeat from being counted twice. The third rule is a prominence check, which measures how far the peak stands above the nearby valleys and rejects anything that does not stand out enough to be a real beat. If a peak passes all three rules it gets added to an array.

Once at least two peaks have been collected the program calculates the BPM. It does this by measuring the spacing between each pair of consecutive peaks, then taking the median of those spacings rather than the simple average so a single bad reading cannot throw the result off. That spacing gets converted from samples into seconds, and from seconds into beats per minute.

The product also shows emoticons depending on what the machine sees, a heart if it has a valid BPM and a skull if no pulse has been found for more than half a second. This is all done through an LCD screen which also displays the current BPM number underneath the icon. An example of what a user may see is shown below.

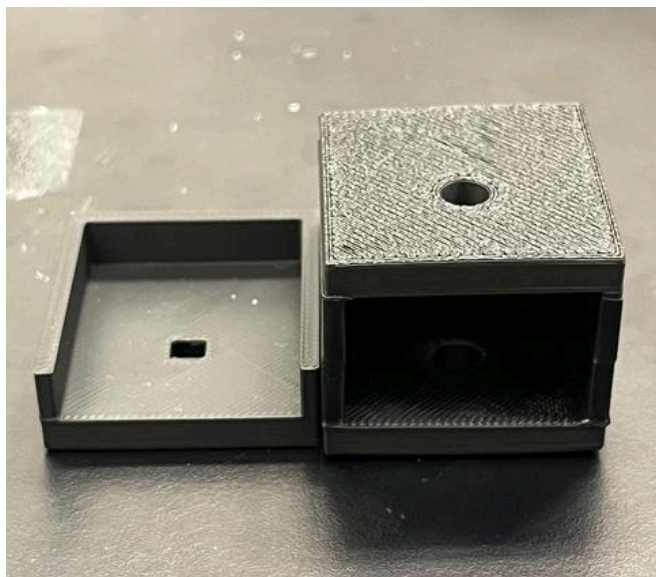
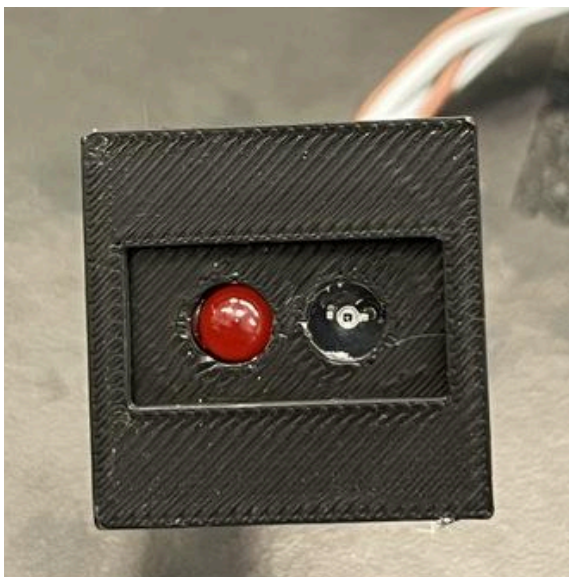
A close-up of a blue LCD screen displaying the text "Pulseometer" followed by a small icon of a heart with a pulse line, and "BPM: 72.5" below it.A close-up of a blue LCD screen displaying the text "Pulseometer" followed by a small icon of a heart with a pulse line, and "BPM: ---" below it.

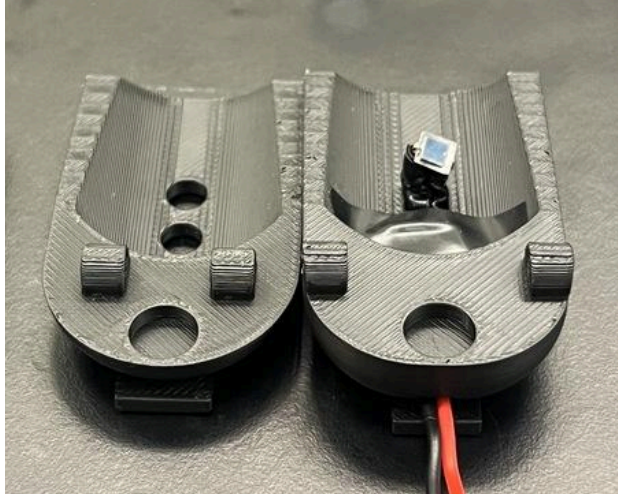
There was a moment when creating the code that it ran in python and the arduino had a code that took the data and sent it to a laptop. This was supposed to make it easier to conduct signal processing; however when we decided we wanted to have the device stand alone, this meant reverting all the python code back into arduino code to handle the processing. Another change within the code since the first edition, there was initially a Savitzky-Golay filter that was removed. The filter was supposed to be a way to reduce any noise so peak detection becomes easier; however the filter was filtering multiple peaks into one. This made the data completely inaccurate, so we decided to abandon the filter. Besides that, the code has stayed relatively the same.

Link to view the code: [C.R.O.I Pulsometer Code](#)

Product Design

The final design of the C.R.O.I pulseometer is a traditional finger clip that latches on the pointer finger or thumb. This finger clip has an LED protruding into the interior of the clip that rests against the users finger, with a photodiode sunken into the casing of the clip directly opposite the LED, such that the light shines through the users finger. This finger clip was the creation of many rounds of prototyping, starting from a “watch” design that used reflective PPG instead of transmissive, then to a “finger box” design that had poor optical isolation and ergonomics, eventually moving on to the finger clip, with many iterations of design to get it ergonomical and effective. All of these designs were CAD-ed with autodesk fusion in-house, with inspiration taken from on-market pulseometers. The casing for the design was made such that it could fit all the circuit components and the arduino while having a viewing port for the LCD screen. For references to all the above see the following photo gallery.



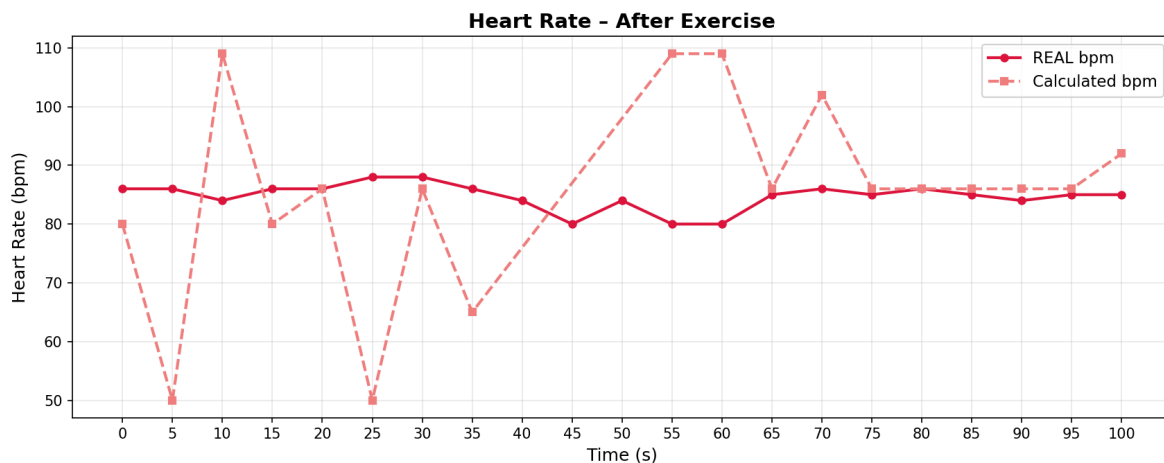
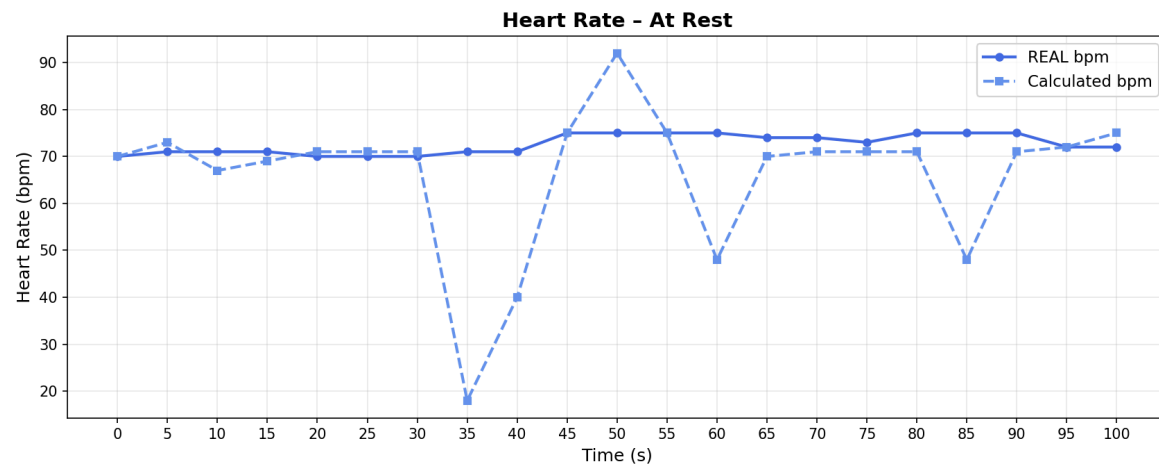




Performance Data

The performance of the pulseometer is highly dependent on how it is worn. We found that, when it is worn correctly, it can give a quite accurate readout of the users bpm. However, wearing it “correctly” is often quite difficult and finicky. If the photodiode is too covered, or not covered enough, by the user's finger, then the signal goes to zero. Similarly, if the user is moving at all, or has not positioned their finger in just the right way, the signal goes to zero. This finicky nature of the pulseometer would obviously present a challenge in bringing it to market, but ideally with improved manufacturing methods (injection molding the casing, using foam or other soft materials to grip the finger better) these problems could be solved. Given these drawbacks, we tested the pulseometer against a commercial baseline: an apple watch. We collected theoretical bpm data from our pulseometer and “true” bpm data from the apple watch, taking one data point

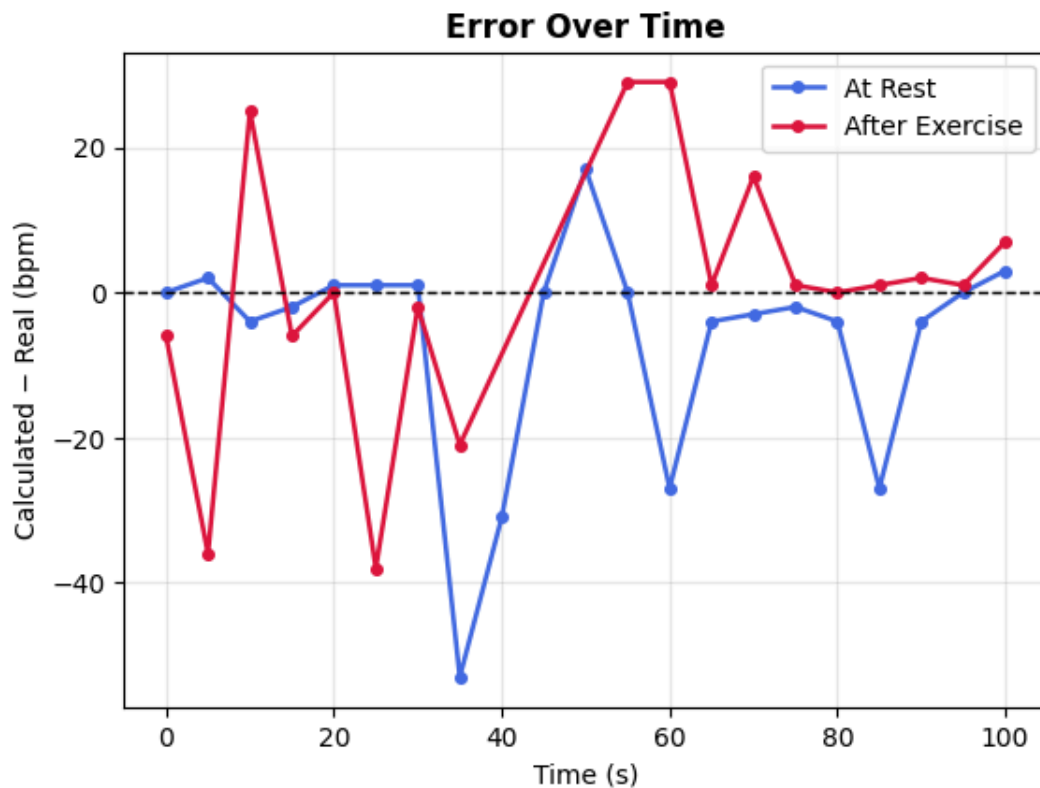
every 5 seconds for 100 seconds total. We did this while at rest, then after a brief workout, to compare the difference in product performance. See below.



One can see there are clearly areas in both graphs where the pulseometer and apple watch agree, but this is interspersed with large deviations in behavior from our pulseometer. The “after exercise” data clearly show a drop in performance from the “at rest” data, most likely due to the fact that the users hands were dirtier, sweatier, and less stable in the pulseometer than in the at rest data set.

We quantified the error in these data sets by finding the mean absolute error (MAE) across the 100 second time frame. We found that the at rest data had an MAE of 8.9bpm, while the after exercise data had an MAE of 12.3bpm, where most of this error comes from the large spikes and

drops in the pulseometer, created by slight change in the users movement. We visualized this error by showing the value calculated-real bpm over time.



Future Considerations

One of the main areas for improvement is the peak detection algorithm. The current implementation uses a simple set of three rules which works reasonably well under ideal conditions, but struggles when the signal gets noisy or the user moves. Future iterations would benefit from researching more advanced peak detection methods, and moving beyond arduino in order to do this more advanced signal processing.

Another goal is to make the device fully wireless and wearable, removing the need for the user to be tethered to the larger box that currently holds the Arduino, circuitry, and LCD screen. The ideal final form would be a self-contained finger clip that houses everything inside, with bluetooth sending the BPM reading to a phone or computer instead of relying on an onboard display.

The third major addition would be blood oxygen saturation (SpO2) measurement, turning the pulseometer into a true pulse oximeter. SpO2 is calculated by comparing how the finger absorbs two different wavelengths of light, typically infrared and red. Implementing this would require adding a second red LED next to the existing IR LED, alternating between the two during sampling, and processing the ratio of their absorbances in real time. With both BPM and SpO2 in

one device, our product would be directly comparable to the commercial finger clip oximeters already on the market.

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